AYMANE AIT DADS

Data Science Intern - Model Optimization | Neural Network Inference · ML Pipeline Engineering · Edge AI

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, $106K+ prize pool) by designing and executing an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer with BF16 mixed precision and 8,192-token context length. Reduced manual network analysis time by 60% at Orange Maroc by building production-grade Random Forest and K-Means pipelines on 100K+ records, directly informing network optimization decisions. Brings hands-on expertise in model fine-tuning, ablation-driven optimization, neural network inference pipelines, and ML framework engineering - skills directly applicable to quadric's co-optimized hardware-software inference stack targeting edge and endpoint devices. Aims to contribute to efficient neural network graph optimization and model performance benchmarking across quadric's GPNPU architecture.

CORE COMPETENCIES

Neural Network Inference | Model Optimization | Edge AI Deployment | Ablation Studies | ML Pipeline Engineering | Mixed Precision Training | Performance Benchmarking | Feature Engineering

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a **Random Forest + K-Means inference pipeline** on 100K+ fiber-optic network records, classifying network quality across thousands of nodes using upload speed, latency, and jitter features.
- Developed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **60%** through automated feature engineering, model evaluation workflows, and reproducible Python-based data processing scripts using Pandas and Scikit-learn.
- Delivered stakeholder presentations translating **model optimization outputs** into actionable maintenance recommendations, bridging technical results and business decision-making using Power BI dashboards.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge — *Kaggle Competition · In Progress (Jun 2026) · Ranked 17th / 3,000+ Teams*

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context), targeting all attention projections, MLP layers, and lm_head for maximum adapter coverage.
- Ran **controlled ablation experiments** across learning rate schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers yielding 605 verified cipher rows and 64 verified bit-manipulation rows for clean chain-of-thought training data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton · Kaggle GPU

AI-Generated Image Detection — *EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · 1st in Class Leaderboard*

- Ranked **1st in the EURECOM class private Kaggle leaderboard** (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M parameters) with a custom MLP head and dual learning rates (2e-6 backbone, 5e-4 head), trained on 250K images across 25+ generator types with FP16 mixed precision.
- Submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; identified through ablation that 1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC), confirming out-of-distribution generalization risk from over-fitting on synthetic image distributions.

PyTorch · OpenCLIP · Hugging Face Transformers · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM — *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Image Security, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya — *2021 - 2023*

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | Hugging Face Transformers | Unsloth | TRL | PEFT | vLLM

Model Optimization & Training: LoRA | QLoRA | BF16/FP16 Mixed Precision | Ablation Studies | Ensemble Methods | Feature Engineering | CUDA / Triton

Data Science & Infrastructure: Python | SQL | Pandas | NumPy | Power BI | Git | Docker (basic) | Linux